

Switching Basics Lab Report

Task 1: Open Cisco Packet Tracer or any network simulator, create a simple network with one switch and two PCs, and use the 'show mac address-table' command to observe how the MAC address table is initially empty.

A simple topology was built in Cisco Packet Tracer using one 2960-24TT switch and two PCs, PC0 (10.0.0.2) connected to port Fa0/1 and PC1 (10.0.0.3) connected to port Fa0/2.

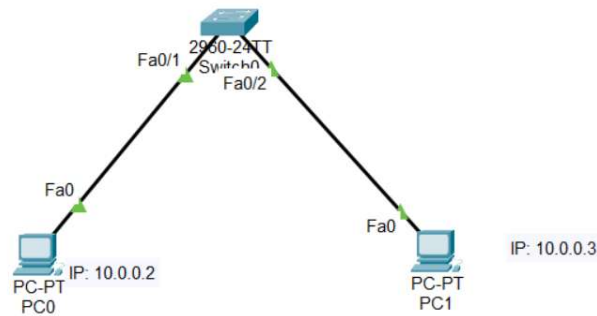


Figure 1: Initial topology — Switch with PC0 and PC1

Before any traffic was sent, the command `show mac address-table` was run on the switch. The table came back completely empty, since the switch had not yet seen any frames from either PC and therefore had no MAC-to-port mappings recorded.

```
Switch#show mac ?
  address-table  MAC forwarding table
Switch#show mac address-table ?
  dynamic       dynamic entry type
  interfaces     interface entry type
  static         static entry type
  <cr>
Switch#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
Switch#
```

Figure 2: MAC address table is empty before any communication

Task 2: Send a ping from PC1 to PC2 in your simulated network, then immediately run 'show mac address-table' on the switch and note how the MAC addresses of both PCs appear in the table. Hint: Try pinging in both directions to see how the table updates.

Next, PC0 was made to ping PC1 (10.0.0.3). The ping completed successfully with 0% packet loss, confirming that PC0 and PC1 could reach each other through the switch.

```

PC0
Physical | Config | Desktop | Programming | Attributes |
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.0.3

Pinging 10.0.0.3 with 32 bytes of data:

Reply from 10.0.0.3: bytes=32 time<1ms TTL=128
Reply from 10.0.0.3: bytes=32 time<1ms TTL=128
Reply from 10.0.0.3: bytes=32 time<1ms TTL=128
Reply from 10.0.0.3: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

```

Figure 3: Successful ping from PC0 to PC1 (10.0.0.3)

Immediately after the ping, show mac address-table was run again on the switch. This time the table showed two dynamic entries — one for each PC's MAC address, mapped to the port it was plugged into.

```

Switch#show mac address-table
          Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0001.c701.d424    DYNAMIC Fa0/2
1       0007.ece7.c9eb    DYNAMIC Fa0/1
Switch#

```

Figure 4: MAC address table after PC0 ↔ PC1 ping

Port	MAC Address
Fa0/1	0007.ece7.c9eb (PC1 / 10.0.0.3)
Fa0/2	0001.c701.d424 (PC0 / 10.0.0.2)

This happened because the switch learns MAC addresses by looking at the source address of every frame it receives on a port. When PC0 sent the ICMP echo request, the switch recorded PC0's MAC on Fa0/2. When PC1 replied, the switch recorded PC1's MAC on Fa0/1. Pinging in both directions is what let the switch learn both addresses in a single exchange.

Task 3: Add a third PC to your network, connect it to the switch, and ping from PC3 to PC1. Observe and record how the MAC address table changes after this communication.

A third PC, PC2 (10.0.0.4), was added and connected to switch port Fa0/3. PC2 was then used to ping PC0 (10.0.0.2).

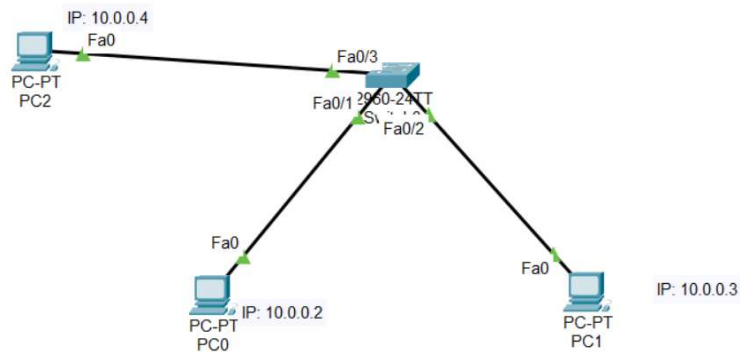


Figure 5: Updated topology with PC2 added on Fa0/3

After the ping, the MAC address table now contained a third dynamic entry for PC2's MAC address on Fa0/3, in addition to the two entries already learned earlier. The existing entries for PC0 and PC1 remained unchanged — the switch simply added the new mapping rather than replacing anything.

```
Switch#show mac address-table
Mac Address Table
-----
Vlan    Mac Address      Type        Ports
----    -
1       0001.c701.d424   DYNAMIC     Fa0/2
1       0007.ece7.c9eb   DYNAMIC     Fa0/1
1       00d0.977b.3841   DYNAMIC     Fa0/3
Switch#
```

Figure 6: MAC address table after PC2 joins and pings PC0

Port	MAC Address
Fa0/1	0007.ece7.c9eb (PC1)
Fa0/2	0001.c701.d424 (PC0)
Fa0/3	00d0.977b.3841 (PC2)

This shows that the MAC address table builds up incrementally: every time a new device sends a frame through the switch, its MAC address and the port it entered on get added to the table. The table never had to relearn PC0 or PC1 — it only added the one new entry it hadn't seen before.

Task 4: Suppose you see the following MAC address table on a switch: Port 1 - 00:11:22:33:44:55, Port 2 - 66:77:88:99:AA:BB. If a frame arrives at the switch on Port 1 with destination MAC 66:77:88:99:AA:BB, explain step-by-step how the switch will forward this frame.

Given switch MAC address table:

Port	MAC Address
Port 1	00:11:22:33:44:55
Port 2	66:77:88:99:AA:BB

Scenario: A frame arrives at the switch on Port 1, with a destination MAC address of 66:77:88:99:AA:BB. Here is what the switch does, step by step:

- Step 1 – Learn the source: The switch first looks at the frame's source MAC address (00:11:22:33:44:55) and checks whether it is already in the table on Port 1. It is, so no new entry is needed (if it were new, the switch would add it now).
- Step 2 – Look up the destination: The switch takes the frame's destination MAC address, 66:77:88:99:AA:BB, and searches its MAC address table for a matching entry.
- Step 3 – Match found: The table shows that 66:77:88:99:AA:BB is reachable out Port 2.
- Step 4 – Forward, not flood: Because a match was found, the switch does not broadcast the frame out every port. Instead it forwards the frame only out Port 2, straight to the device that owns that MAC address.
- Step 5 – Delivery: The device connected to Port 2 receives the frame. All other ports on the switch, including Port 1 where the frame arrived, are not sent a copy.

If the destination MAC address had not been found in the table, the switch would have flooded the frame out every port except the one it arrived on, and learned the reply's source address once the destination device responded.

Task 5: Draw a diagram showing how a switch learns and uses MAC addresses to forward frames in a network with 4 devices, and write a short explanation (3-4 sentences) describing the process in your own words. Constraint: Do not copy definitions from the textbook; use your own understanding and examples.

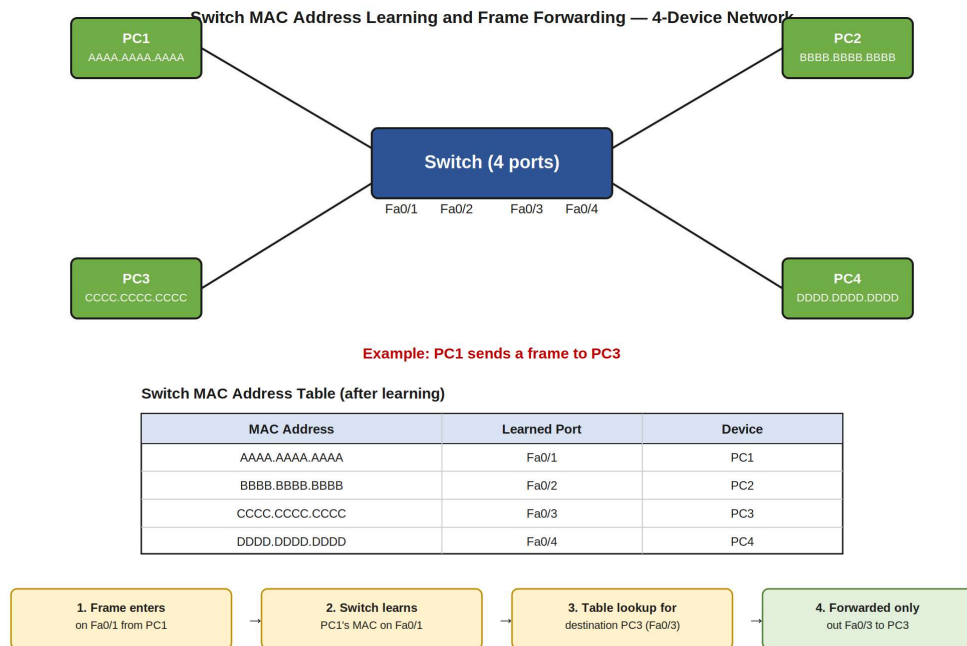


Figure 7: A switch learning MAC addresses from four PCs and forwarding a frame from PC1 to PC3

A switch starts out not knowing where any device is. As each PC sends traffic, the switch takes a look at the source MAC address of the incoming frame and notes down which physical port that address came in on, and it happens automatically without any configuration.

Once all four PCs in the diagram have sent at least one frame, the switch has a complete map of MAC address to port for the whole network. After that, when a frame comes in asking for a specific destination MAC address, the switch checks its table and sends the frame out only the one port that leads to that device, rather than repeating it out every port the way a hub would. This is why, in the example above, a

frame from PC1 addressed to PC3 goes straight out Fa0/3 and PC2 and PC4 never see it at all — the switch keeps traffic contained to just the two ports that actually need it.